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Belt conveyor idler fault detection algorithm based on improved YOLOv5

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The rapid expansion of the coal mining industry has introduced significant safety risks, particularly within the harsh environments of open-pit coal mines. The safe and stable operation of belt conveyor idlers is crucial not only for ensuring efficient coal production but also for safeguarding the lives of coal mine workers. Therefore, this paper proposes a method based on deep learning for real-time detection of conveyor idler faults. The selected YOLOv5 network is analyzed and improved based on the training results. First, the coordinate attention mechanism is integrated into the model to reassign the weights across different channels. Subsequently, the α-CIoU localization loss function replaces the traditional CIoU to enhance the model's regression accuracy. Experimental results demonstrate that the enhanced YOLOv5 algorithm achieves a 95.3% mAP on the self-constructed infrared image dataset, surpassing the original algorithm by 2.7%. Moreover, with a processing speed of 285 FPS, it accurately performs the defect detection of conveyor idlers while satisfying real-time operational requirements.

Keywords Belt conveyors, Idler, YOLOv5, Attention mechanism, α-CIoU

Thermal power serves as the foremost electricity source in China at present. By capitalizing on its coal resource advantages, the Zhundong region in Xinjiang has become a pivotal hub for the "West-to-East Power Transmission" initiative, in line with the national objective of transferring energy resources from the western to the eastern regions of China. Remarkably, the Changji-Guquan ± 1100 kV Ultra High Voltage Direct Current (UHVDC) transmission project has transmitted over 20 billion kilowatt-hours of electricity, actively participating in the national initiative of west-to-east power transmission. Ensuring a stable coal supply is paramount for maintaining electricity production.

Belt conveyors are widely used in coal transportation due to their large capacity, low power consumption, simple structure, low cost, and high reliability. The stability and reliability of the conveyor belt drive system are of significant importance. Idlers, as shown in Fig. 1, are the main components that support the conveyor belt along its length and typically comprise rollers, shafts, bearings, rack, and sealing components. Due to their prolonged operation under heavy loads, idlers are prone to jamming faults, leading to excessive localized temperatures and even fire incidents, resulting in significant economic losses and human casualties. To ensure economic benefits and personnel safety, it is important to conduct regular inspections of the idlers to detect any harmful changes in their operational condition, allowing for repairs or replacements when necessary. However, due to the large number and dispersed distribution of idlers, dynamic load forces, and complex working environments, detecting faults in idlers is extremely challenging.

Common types of idler failures include idler bearing damage, idler shell wear, and idler jamming. Idler bearing damage can result from mismatches between idlers and conveyor belts, poor idler shaft sealing leading to lubricant oxidation, and inadequate or poor-quality lubricant filling in idler assemblies. Idler shell wear may stem from excessive rotational resistance, misalignment between idler rotation and belt movement causing friction deviation, and corrosive environments accelerating idler deterioration. Idler jamming can be caused by lubricant failure or foreign objects wedging. Currently, the methods for detecting idler faults primarily include manual inspections, acoustic signal analysis, vibration analysis, and temperature analysis. Manual inspection involves maintenance personnel using tools to examine the idlers on conveyor belts. Due to coal mine conveyor systems can extend for several kilometers, conducting a complete inspection requires significant manpower and time, resulting in a heavy workload. This also poses a risk of injury to inspection personnel during the process. Acoustic signal analysis and vibration analysis primarily involve feature extraction and noise reduction of sound or vibration signals, followed by classification using machine learning methods[1–4]. However, this type of method has the disadvantages of poor reliability, high cost, and difficult implementation. First, since the vibration sensors

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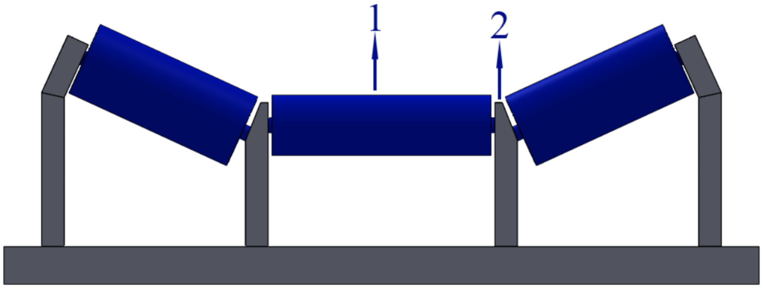

Fig. 1. The structural diagram of the belt conveyor idler. 1-roller; 2-rack.

must be directly or indirectly mounted on the target idlers, and the vibration of the frame, conveyor belt, and other rotating components will interfere with the monitoring data, which makes this type of method unreliable. Second, the number of idlers is large and the distribution is scattered, many sensors are required, which results in high hardware, construction, and maintenance costs. Finally, due to the small size of the idler, it is difficult to mount the contact sensors, and the data transmission and power supply of the sensors are difficult to effectively solve and implement[5]. Furthermore, the frequent occurrence of adverse weather conditions such as strong winds and heavy fog in Xinjiang, China, significantly interferes with acoustic and vibration signals. Consequently, acoustic signal and vibration analysis methods are not suitable for open-pit coal mines in Xinjiang, China.

When an idler fails, its movement becomes irregular and the rotational resistance increases, resulting in localized temperature rise due to friction between the idler and the conveyor belt or within the idler itself. So using infrared image data for detecting idler faults is highly feasible. Preprocessing the data is a fundamental step in utilizing infrared image AI technology to identify idler faults, which can mitigate the impact of noise and complex backgrounds on idler fault recognition[6]. Zhang et al.[7] employed a 3D denoising algorithm to suppress random noise in the data and designed an image enhancement algorithm based on multi-scale retinal layer theory to enhance image details. Pathak et al.[8] utilized discrete wavelet transformation and particle swarm optimization to segment noisy data, achieving satisfactory segmentation; however, further enhancement is needed in preprocessing to better restore image data. Many scholars have studied idler fault recognition methods based on infrared video AI technology. Szrek et al.[9] deployed RGB cameras and infrared cameras on inspection robots, analyzing and processing data from these two devices separately and then merging them to achieve idler fault recognition and localization. Siami et al.[10] proposed an automatic idler segmentation method and shape detection algorithm based on anomalies, comparing segmented hotspots' shapes and positions with RGB images to quantitatively assess conveyor idler states. Szurgacz et al.[11] utilized thermal imaging diagnostic methods to measure abnormal idler infrared radiation, achieving idler fault diagnosis under conveyor belt operation conditions. Ma et al.[12] utilized infrared data processing techniques for idler area segmentation and feature extraction, making fault level judgments and issuing fault warnings based on temperature. Liu et al.[13] proposed a conveyor belt idler fault analysis method based on sound and thermal infrared features, enabling the identification and display of idler fault severity. CARVALHO R et al.[14] used infrared cameras to collect data, converting infrared images into temperature value matrices to obtain idler temperatures and detect faults. DABEK P et al.[15] extracted ROI from RGB data, identified hotspots in infrared data using hotspot detection algorithms, and finally merged infrared fault points into RGB data to achieve accurate idler fault recognition. Liu[16] used optical flow extraction for edge detection to ensure video integrity, applying an improved CNN and LSTM dual-stream network model to recognize idler abnormal states from processed videos. Jin[17] focused on idler infrared image component recognition, selecting the fast and lightweight YOLOv5 algorithm for rapid and accurate component recognition. The model files were integrated into MATLAB programs, enabling model conversion and mutual utilization. Experimental results demonstrate that this algorithm achieves high accuracy and speed, meeting practical requirements for lightweight deployment. Inspection robots can be equipped with a thermal infrared imager for regular idler fault detection, reducing hardware and maintenance costs, further enhancing recognition accuracy and speed.

The method for fault identification of idlers based on infrared technology faces the following technical challenges:

(1) Currently, there is no established infrared image dataset specifically for conveyor belt idlers.
(2) The automatic identification of idler needs to be addressed in the application of infrared technology for fault detection.
(3) There are issues with low recognition accuracy and poor real-time performance.

To meet the demand for fault detection of idlers in conveyor belts at open-pit coal mines in Xinjiang, China, this paper proposes an abnormal detection method based on an improved YOLOv5 model. First, an infrared image dataset specifically for conveyor belt idlers is established by collecting data from various scenarios and labeling the dataset accordingly. This abnormal idler infrared dataset is constructed for future research in the field of deep learning. Second, to enhance the model's recognition capability for idlers, the CA attention mechanism is integrated into the original YOLOv5 structure, and the traditional CIoU loss function is replaced with the α-CIoU loss function to improve regression accuracy, significantly enhancing the overall recognition precision of the experiments. The experimental results indicate that the abnormal detection model for conveyor belt idlers based on the improved YOLOv5 effectively determines whether issues exist with the idler rollers, meeting practical requirements.

Related work

YOLOv5 overview

YOLO is a revolutionary object detection algorithm that achieves rapid recognition and localization of objects through a single forward pass, significantly enhancing detection performance with its end-to-end approach. As the YOLO series algorithms have evolved from YOLOv1 to YOLOv9, their performance has continually improved. Among them, YOLOv5 is currently the most widely applied, known for its high accuracy, speed, and versatility. Compared to YOLOv7 and YOLOv8, YOLOv5 stands out for its lightweight nature and ease of use. It maintains high accuracy while reducing model complexity, providing robust support for various application scenarios. Therefore, in this paper, we comprehensively consider and select the YOLOv5 model.

The YOLOv5 algorithm consists of four models: YOLOv5s, YOLOv5m, YOLOv5l, and YOLOv5x. YOLOv5 incorporates various modules such as the Feature Pyramid Network (FPN) and Spatial Pyramid Pooling Faster (SPPF) within the convolutional neural network architecture, which can be segmented into four primary components: Input, Backbone, Neck, and Head.

(1) Input: YOLOv5 employs Mosaic data augmentation at the input stage, where four random images are selected and combined with random scaling, rotation, and concatenation to generate complex training samples. Additionally, techniques such as adaptive anchor calculation and adaptive image scaling are utilized to enhance model generalization and detection accuracy.
(2) Backbone: The YOLOv5 algorithm replaces the previous Focus structure with convolutions and adopts a Cross Stage Partial (CSP) structure in the backbone network. These structures effectively extract features from input images and fuse features from different scales, thereby enhancing the model's feature representation capability.
(3) Neck: YOLOv5 incorporates the Spatial Pyramid Pooling with FPN (SPPF) structure in the neck network. This structure applies pooling operations at different scales, providing features with advantages such as invariance and multi-scale representation, while minimizing information loss.
(4) Output: The output layer processes feature maps of sizes 20×20, 40×40, and 80×80 generated by the neck network. Through the use of specific loss functions and non-maximum suppression, YOLOv5 performs predictions for objects of varying sizes.

Loss function

In the YOLOv5 algorithm, the loss function comprises a classification loss, localization loss, and confidence loss, where the localization loss refers to the error between predicted boxes and ground-truth boxes.

Within the network architecture of YOLOv5, the CIoU (Complete IoU) loss is employed as the localization loss function. CIoU extends upon DIoU (Distance IoU) by incorporating scale regularization terms, encompassing overlap rate, scale, distance, and penalty terms. This comprehensive formulation stabilizes the regression of target boxes, preventing divergence during training. The formula is shown in Eq. (1):

$$L_{CIoU} = 1 - IoU + \frac{\rho^2\left(b, b^{gt}\right)}{c^2} + \beta\nu \tag{1}$$

where $IoU = \frac{A\cap B}{A\cup B}$, $\beta = \frac{\nu}{(1-IoU)+\nu}$, $\nu = \frac{4}{\pi^2}\left(\arctan\frac{w^{gt}}{h^{gt}} - \arctan\frac{w}{h}\right)^2$.

IoU denotes to the ratio of the intersection area between a predicted bounding box and a ground truth bounding box to their union. A and B denote the prediction frame and the real frame. b and b^{gt} denote the centroids of the prediction frame and the real frame. Respectively, ρ denotes the Euclidean distance between the two centroids of the prediction frame and the real frame. c denotes the square of the diagonal length of the minimum enclosing frame. β is the weight factor,ν denotes the similarity of the aspect ratio. w^{gt} denotes the width of the labeled frame. h^{gt} denotes the height of the labeled frame. w denotes the width of the prediction frame, and h denotes the height of the prediction frame.

The α-CIoU is a novel localization loss function proposed by He et al.[18], designed to enhance the flexibility of detection models in regressing bounding boxes across different scales by adjusting a single power parameter α. Moreover, α-CIoU exhibits stronger robustness against small datasets and noise. The formula is shown in Eq. (2):

$$L_{\alpha-CIoU} = 1 - IoU^{\alpha} + \frac{\rho^{2\alpha}\left(b, b^{gt}\right)}{c^{2\alpha}} + (\beta\nu)^{\alpha} \tag{2}$$

The experimental results indicate that the optimal performance is achieved when $\alpha = 3$.

Attention mechanism

The primary function of attention mechanisms is to enable models to concentrate on specific parts of input data, thereby facilitating more effective information processing. This allows models to focus on task-relevant features or contextual information while disregarding irrelevant or less significant data. In the domain of object detection, the introduction of an attention mechanism notably amplifies the detection performance for small objects. Therefore, this paper compares the effect of three attention mechanisms for idler faults detection. The three attention mechanisms are SeNet, SimAM, and CA attention mechanisms.

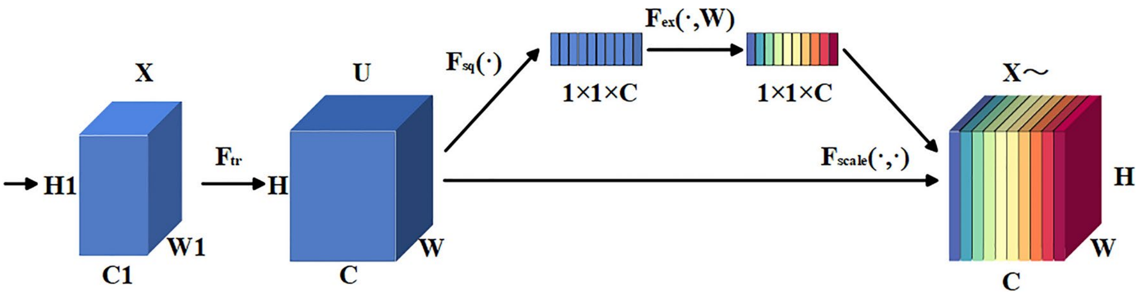

Fig. 2. The structure of SeNet module.

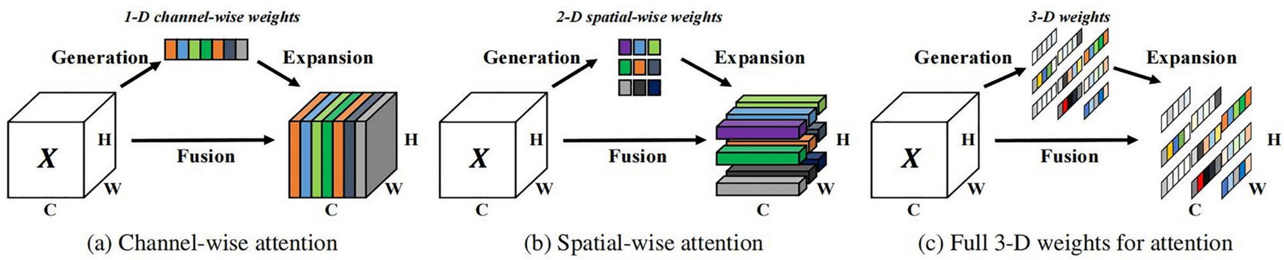

Fig. 3. The structure of SimAM module.

SeNet attention module
SeNet (Squeeze-and-Excitation Networks) attention module[19] was proposed by Hu et al., in 2017, and its network structure is shown in Fig. 2[19]. SeNet is a channel-wise attention mechanism that enhances the network's performance by amplifying its focus on different feature channels. The core idea involves learning an attention weight vector representing the relationships between channels, allowing for the redistribution of information among channels to augment the network's attention towards important feature channels. The SeNet module comprises three components: the squeeze operation, excitation operation, and scale operation.

In the squeeze operation, the input feature map of size H × W × C is compressed into a 1 × 1 × C vector through global average pooling, where C denotes the number of channels in the input feature map. Subsequently, this vector is mapped to a smaller vector through a fully connected layer, yielding the attention weight vector. In the excitation operation, a sigmoid function is applied to compress each element of this vector to a range between 0 and 1, thereby obtaining the attention weights for each channel. The purpose of the excitation operation is to recalibrate features, reduce model parameter complexity, and enhance model generalization. In the scale operation, the 1 × 1 × C attention weight obtained from the excitation operation is multiplied element-wise with the corresponding channels of the original input H × W × C feature map, yielding the weighted feature map.

SimAM attention module
The SimAM (Similarity Attention Mechanism) attention module[20] is a lightweight, parameter-free convolutional neural network (CNN) attention mechanism, and its network structure is shown in Fig. 3[20]. In object detection tasks, the SimAM attention mechanism can assist the model in better focusing on regions of interest, thereby enhancing the accuracy and robustness of object detection.

The SimAM attention mechanism first extracts feature maps from images using CNN. Subsequently, it calculates the similarity between each channel of the feature maps. Then, based on the computed similarities, SimAM generates an attention weight map, which indicates the importance of each channel or feature map for the final output. Finally, the attention weight map is multiplied element-wise with the original feature maps to obtain the weighted feature maps.

CA attention module
CA (Coordinate Attention) attention module[21] is an attention module proposed by Hou et al. It is a targeted improvement based on the SeNet module, and its structural diagram is illustrated in Fig. 4[21]. In contrast to the SeNet module, the CA module embeds positional information into channel attention, allowing attention to be considered not only in the channel dimension but also in the spatial dimension, as illustrated in the figure.

The Coordinate Attention (CA) mechanism first performs global average pooling separately along the height and width dimensions of the feature map, as shown in Eqs. (3) and (4).

$$z_c^h(h) = \frac{1}{W} \sum_{0 \le i < W} x_i(h, j) \tag{3}$$

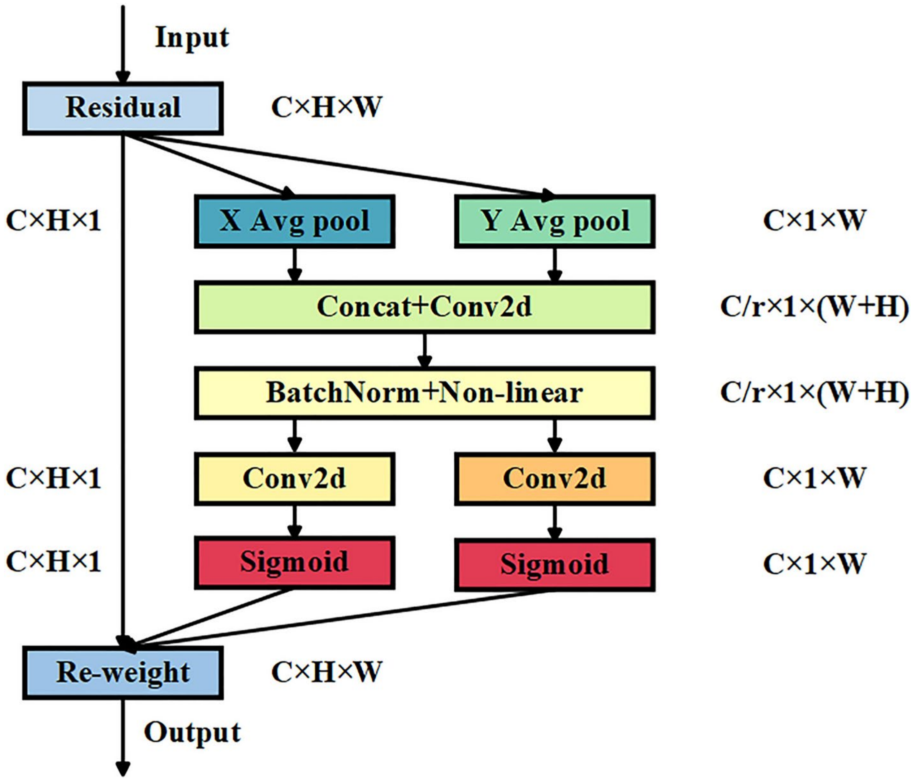

Fig. 4. The structure of CA module.

$$z_c^w\left(w\right)=\frac{1}{H}\sum_{0\le j<H}x_c\left(j,w\right) \tag{4}$$

Next, the feature map is concatenated and processed with a 1 × 1 convolutional kernel for dimension reduction. Subsequently, an activation operation is applied, followed by a split operation along the spatial dimension. This is followed by combining with a sigmoid activation function to obtain the final attention vectors g_c^h and g_c^w . By utilizing parameters i and j, specific positions $x_c(i,j)$ in the input image can be located. Multiplying the corresponding position $x_c(i,j)$ by the weights yields the output feature map, as shown in Eq.(5).

$$y_c\left(i,j\right)=x_c\left(i,j\right)\times g_c^h\left(i\right)\times g_c^w\left(j\right) \tag{5}$$

Methodology
Improved the YOLO v5 model
To further enhance feature extraction, an attention mechanism module was added after the 6th layer of the backbone network. The optimized network structure is depicted in the Fig. 5 below.

Fault detection processing flow
This paper presents a method for detecting faults in flywheels based on infrared technology, utilizing hardware devices such as an infrared thermal imaging camera and a wireless transmission system for data acquisition and transmission. The received data is processed using an improved YOLOv5 model for fault diagnosis. If the “gw” tag is detected, it indicates a fault, triggering a fault alarm. The fault detection processing flowchart is shown in the Fig. 6.

Experiments and analysis
Data sets and experimental settings
The quality of the data set is crucial for the results of deep learning. Currently, there is no publicly available dataset of infrared images for detecting idlers abnormalities. This paper opts to construct its own dataset, comprising 3058 images sourced from the thermal imaging camera of an inspection robot deployed by an energy company in Xinjiang, China. To expand the dataset of abnormal high-temperature images, augmentation techniques such as rotation and mirroring are employed, resulting in a total of 4500 images. The ratio of normal to abnormal high-temperature images is maintained at 1:1. The abnormal high-temperature images are annotated using the labelImg annotation tool, with the category labeled as “gw” and saved in the YOLO series’ txt format. The dataset is shuffled and divided into training and testing sets in an 8:2 ratio, yielding 3600 training images and 900 testing images. An example of the dataset is illustrated in the Fig. 7. The size distribution of all bounding boxes in the

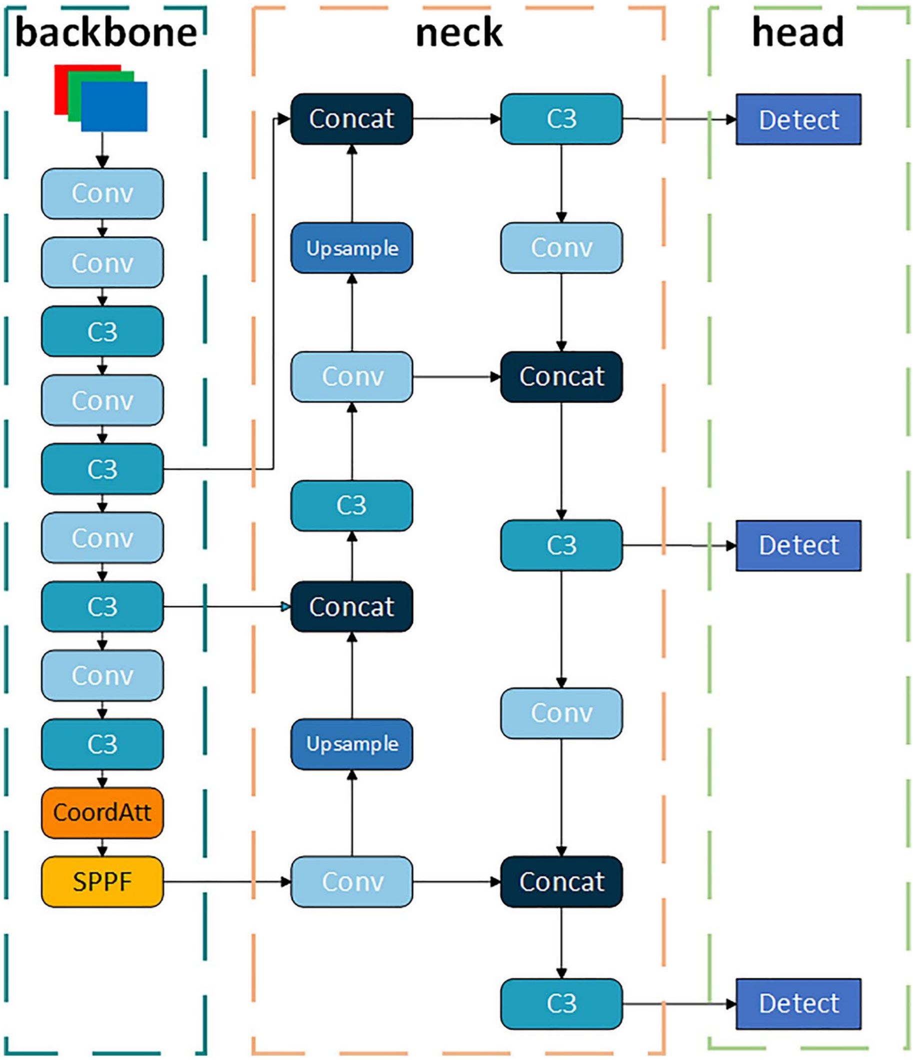

Fig. 5. Improved YOLOv5s model.

training set is depicted in Fig. 8, where the horizontal and vertical axes represent the width and height of the bounding boxes, respectively. In the figure, deep blue indicates a higher frequency of labels within that region.

The experimental platform operates on Windows 11 Professional Edition 22H2, with a CPU featuring the 13th Gen Intel(R) Core(TM) i7-13700F@2.10 GHz processor. The graphics card utilized is the NVIDIA GeForce RTX 4060Ti, with 8 GB of VRAM, and the system is equipped with 16 GB of RAM. The entire experimental framework is based on PyTorch 2.0.0, within a Python 3.10 environment, and GPU acceleration software includes CUDA 12.1 and CuDNN 8.9.2. During experimentation, training parameters are set as follows: a batch size of 16, with 4 workers, and the number of epochs is set to 300.

Indicators of evaluation

In order to assess the performance of the improved YOLOv5 algorithm, precision (P), recall (R), average precision (AP), mean average precision (mAP), and detection speed (frames per second, FPS) are adopted as performance evaluation metrics. The calculation can be calculated using the following equations.

$$P = \frac{TP}{TP+FP} \tag{6}$$

$$R = \frac{TP}{TP+FN} \tag{7}$$

$$P = \int_{0}^{1} P\left(R\right) \tag{8}$$

$$mAP = \frac{\sum_{j=1}^{c} AP_j}{c} \tag{9}$$

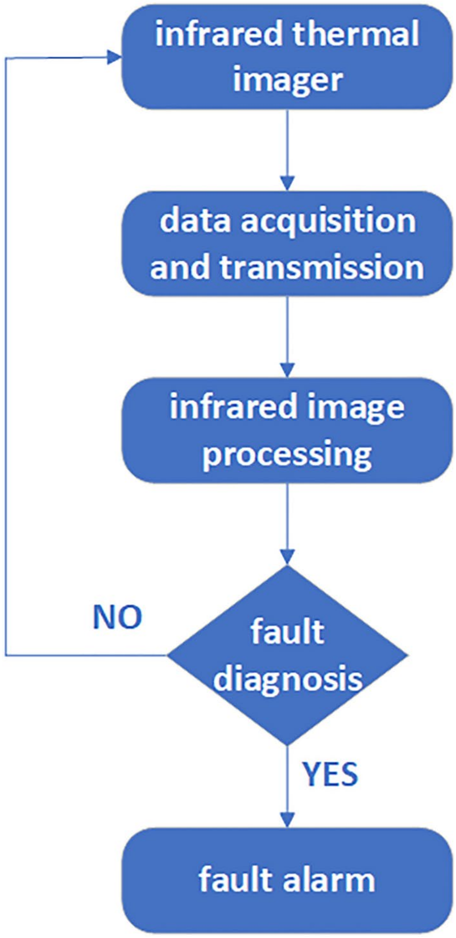

Fig. 6. Fault detection processing flowchart.

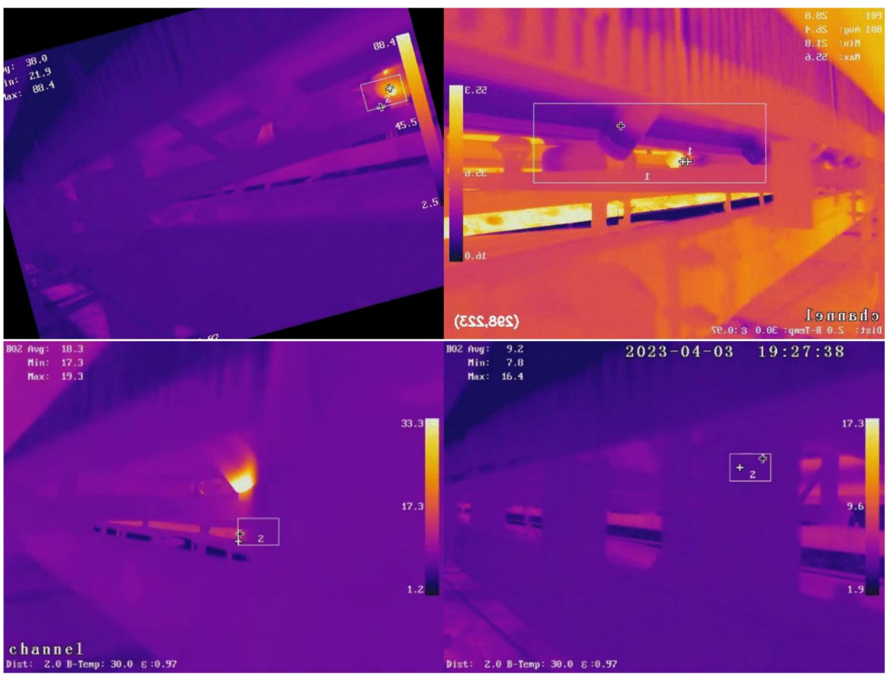

Fig. 7. Dataset example graph.

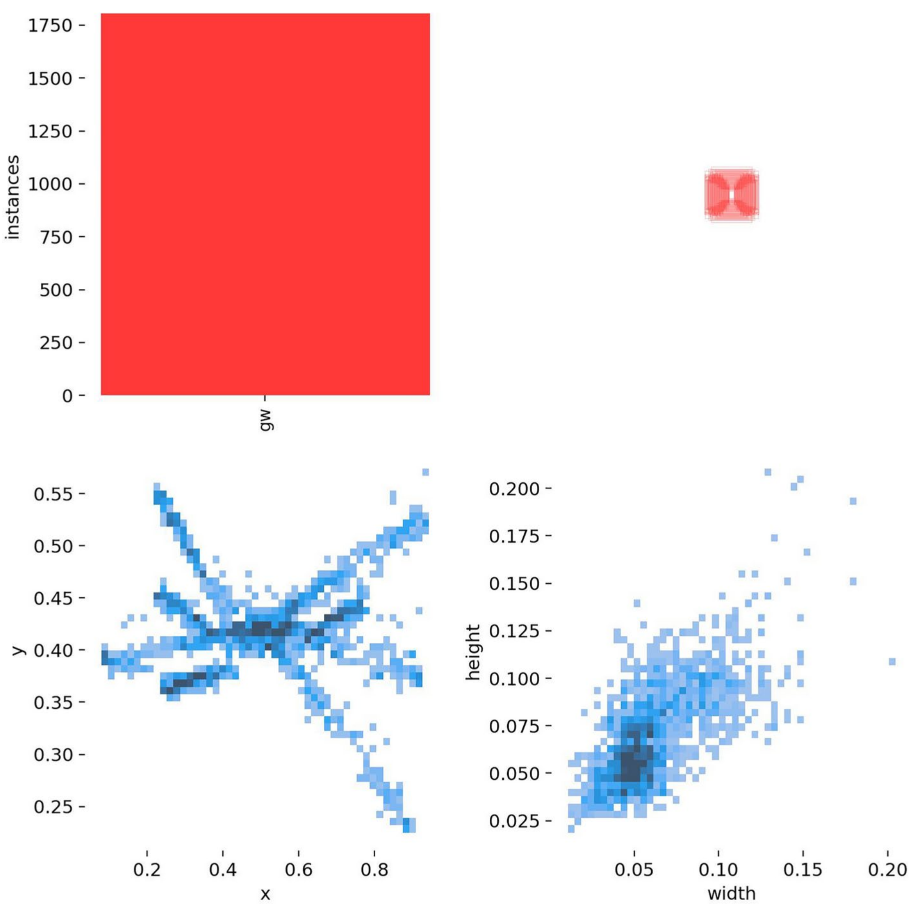

Fig. 8. Label box size distribution.

In the equations, TP represents the number of true positive predictions, FP represents the number of false positive predictions, and FN represents the number of false negative predictions. P is the ratio of correct positive predictions among all predictions classified as positive, while R is the ratio of correctly predicted positive samples among all positive samples. AP represents the area under the Precision-Recall curve, reflecting the model's performance in predicting positive samples accurately for the current class. mAP is divided into mAP@0.5 and mAP@0.5:0.95 based on IoU thresholds, where 'm' denotes average, and the number after '@' indicates the threshold for determining positive and negative samples based on IoU; samples with IoU greater than the threshold are considered positive, otherwise negative. @0.5:0.95 signifies the threshold range from 0.5 to 0.95. A higher mAP value indicates better network performance. FPS, on the other hand, reflects the number of images the model can process per second.

Ablation experiment

Ablation experiments are conducted on the same dataset for each improved module of YOLOv5 to test the effect of each improved module. The experiments are based on YOLOv5 with the replacement of the loss function with the α-CIoU loss function, and the introduction of the CA attention mechanism, and + indicates that an improvement of a module is added, as shown in Table 1.

The original YOLOv5 model demonstrates mAP of 92.6%, with a precision rate of 91.2% and a recall rate of 94.8%. The YOLOv5s model, augmented with the Coordinate Attention (CA) mechanism, demonstrated substantial enhancements in precision, recall, mAP@0.5, and mAP@0.5:0.95. Substituting the original loss function with the α-CIoU loss function caused a slight decline in recall, but markedly improved precision, mAP@0.5, and mAP@0.5:0.95. As depicted in Table 1, the YOLOv5s + CA combination attained an precision of 93.6%, a recall of 95.6%, an mAP@0.5 of 94.9%, and an mAP@0.5:0.95 of 41.9. Although the YOLOv5s + SimAM combination surpassed YOLOv5s + CA regarding recall and mAP@0.5:0.95, it exhibited significantly lower precision and mAP@0.5. Furthermore, substituting the original loss function with the α-CIoU loss function in the YOLOv5s + SimAM + α-CIoU combination led to a reduction in recall relative to YOLOv5s + SimAM. Considering that our real-time detection project necessitates higher precision and mAP@0.5, the proposed algorithm is overall the optimal choice. In comparison to the original model, our algorithm enhanced precision by 2.5%, recall by 0.5%, mAP@0.5 by 2.7%, and mAP@0.5:0.95 by 1.5%.

To assess the impact of substituting CIoU with α-CIoU in YOLOv5s, we performed a comparative experiment, scrutinizing the model's performance metrics across varying values of α. The results depicted in Table 2 reveal

Model	Precision/%	Recall/%	map@0.5/%	map@0.5:0.95/%
YOLOv5s	91.2	94.8	92.6	41
YOLOv5s + SE	92.1	93.8	93.8	41.7
YOLOv5s + SimAM	92.9	96	94.8	42.7
YOLOv5s + CA	93.6	95.6	94.9	41.9
YOLOv5s + α-CIoU	91.6	94.4	94.6	41.6
YOLOv5s + SE + α-CIoU	92.5	94.7	94.6	41.9
YOLOv5s + SimAM + α-CIoU	92.9	95.3	95.1	42.2
Ours(YOLOv5s + CA + α-CIoU)	93.7	95.3	95.3	42.5

Table 1. Results of ablation experiments.

α	Precision/%	Recall/%	map@0.5/%	map@0.5:0.95/%
α = 1	93.6	95.6	94.9	41.9
α = 2	93	95.6	95.1	42
α = 3	93.7	95.3	95.3	42.5
α = 4	88.7	90.2	91.6	39.7

Table 2. The results for different values of α.

Model	Precision/(%)	Recall/(%)	mAP@0.5/(%)	FPS
Faster RCNN	84.7	90.1	82.7	43
SSD	79.2	91.3	73.5	260
YOLOv5	91.2	94.8	92.6	263
YOLOV7	90.7	87.5	89.5	240
YOLOv8	90.7	93.4	94.9	284
Ours	93.7	95.3	95.3	285

Table 3. Comparison of experimental results.

that at α = 3, although recall exhibited a slight decline compared to α = 1 and α = 2, the precision, mAP@0.5, and mAP@0.5:0.95 achieved their highest values, reaching 93.7%, 95.3%, and 42.5%, respectively. Consequently, it can be concluded that the model's performance metrics are optimally balanced at α = 3.

Comparison experiments

In order to verify the superiority of the improved algorithm in this article, several popular target detection networks are used for comparative experiments: Faster-CNN, SSD, YOLOv5, YOLOv7 and YOLOv8. The experimental process adheres to the principle of controlling variables, and ensuring consistency in both the experimental software and hardware environments. The evaluation indicators use mAP, R, P, and FPS. The experimental results are shown in Table 3.

As shown in Table 3, our model's mAP on the constructed dataset surpasses that of Faster R-CNN, SSD, YOLOv5, YOLOv7, and YOLOv8 by 12.6%, 21.8%, 2.7%, 2.1%, 5.8%, and 0.4%, respectively, demonstrating superior detection performance. The precision ranks as Faster R-CNN < SSD < YOLOv7 = YOLOv8 < Ours, while the recall order is YOLOv7 < YOLOv8 < Faster R-CNN < YOLOv5 < Ours, confirming that our algorithm satisfies the detection accuracy requirements. In summary, compared to the latest YOLOv8, our improved algorithm features fewer parameters and lower computational complexity, achieving 3%, 1.9%, and 0.4% improvements in precision, recall, and mAP, respectively, with reduced training time. When compared to traditional Faster R-CNN and SSD models, our enhanced algorithm not only significantly reduces the number of parameters, computational load, and training time but also achieves higher precision recall, and mAP. The overall results affirm that the improved YOLOv5 model delivers high precision, recall, and mAP, meeting precision requirements. Furthermore, with the thermal infrared camera transmitting video at 20–40 FPS, our detection speed of 285 FPS is adequate for real-time detection.

Test results

In order to visually validate the detection performance of the enhanced YOLOv5 model, this study evaluated idler conveyor anomalies across various scenarios. A comparative analysis of the evaluation results is presented in Fig. 8.

As illustrated in Fig. 9, each group consists of five images. The first group represents the original images, while the second group depicts the detection outcomes utilizing the original YOLOv5 model. The third group

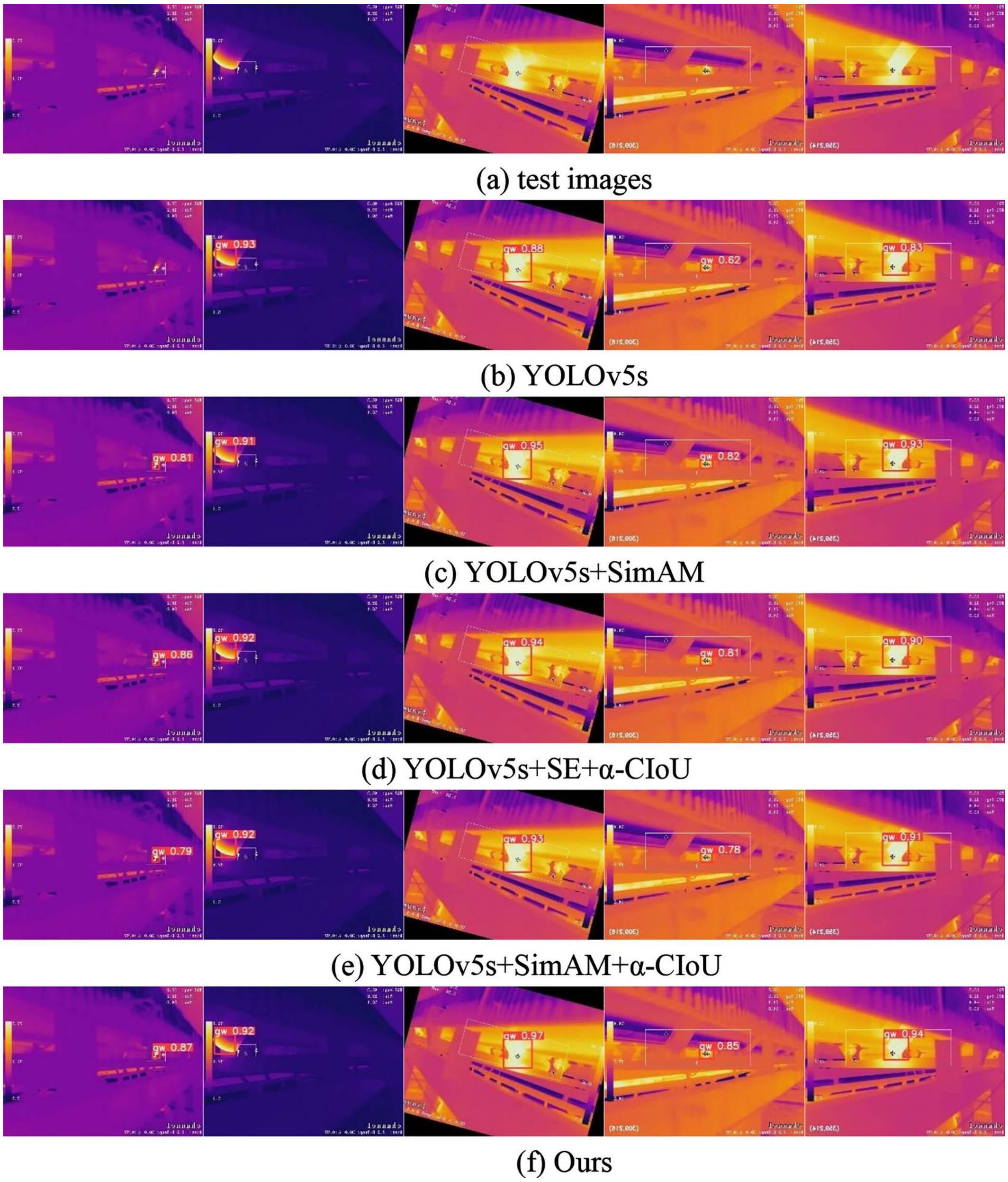

Fig. 9. Comparison of experimental results.

illustrates the detection results obtained with the YOLOv5s + SimAM model, which attained the highest recall and mAP@0.5:0.95 values. The fourth group showcases the detection results leveraging the YOLOv5s + SE + α-CIoU model. The fifth group demonstrates the detection outcomes achieved with the YOLOv5s + SimAM + α-CIoU model. The sixth group exhibits the detection results produced by our proposed improved YOLOv5 model. Our improvement approach incorporates CA mechanism between the Neck and Head, alongside the utilization of the α-CIoU localization loss function to improve regression accuracy. These enhancements aid in reducing information redundancy, mitigating omissions, and erroneous detections in complex scenarios. Experimental results unequivocally demonstrate that our proposed model excels in terms of confidence. Achieving the highest confidence levels in the majority of anomaly detection results. Significantly surpasses the original YOLOv5 model in the detection of small objects. This further substantiates the efficacy of our proposed improved YOLOv5 model.

Conclusion
This paper focuses on enhancing the YOLOv5 model within the framework of deep learning-based object detection. Primarily, the CA attention module is introduced between the Neck and Head regions to augment the network's capability in extracting features related to abnormal targets on belt conveyors idlers. Additionally, the original YOLOv5 loss function is replaced with the α-CIoU loss function, enhancing the model's localization

accuracy and reducing regression errors. Experimental evaluations conducted on the self- constructed dataset reveal a 2.5% increase in precision, a 0.5% increase in recall, and a mean average precision improvement of 2.7% compared to the original YOLOv5s model. The test results proved that the improved algorithm proposed in this study outperforms the original YOLOv5s model, meeting the requirements of idlers defect detection in mining scenarios. The algorithm we propose enables real-time detection in the open-pit coal mines, demonstrating high accuracy. Nonetheless, the limited dataset, constrained by the track limitations of inspection robots, may result in reduced accuracy across varying thermal infrared imaging perspectives. Future research will prioritize the expansion of the dataset, followed by the installation of processing equipment on inspection robots to further enhance real-time performance. Finally, a lightweight model that maintains both accuracy and detection speed will be developed to facilitate the implementation of this model on mobile devices.

Data availability

Data or code presented in this study are available request from the corresponding author.

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Author contributions

Cen Pan designed the study, writed the main manuscript text, collected data, analyzed and interpreted data. Hao Pei, Biao Wang and Wei Liu prepared figures and tables. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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